

NCAA March Madness Prediction System

Mixture-of-Experts, Temporal Modeling, and Prediction Market Trading

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Abstract

We propose a probabilistic prediction system for the NCAA Men’s Basketball Tournament (“March Madness”), targeting both the Kaggle *March Machine Learning Mania* competition (2027 edition) and live trading on the Kalshi prediction market. Our approach combines three methodological innovations over standard baselines: (1) a **Mixture-of-Experts** architecture that routes matchups to specialized sub-models by game type, (2) a **Transformer sequence model** that encodes a team’s full-season trajectory into a dense embedding, and (3) a **market-calibrated ensemble** that fuses model predictions with betting market prices via Bayesian updating. We backtest on 12 years of tournament data (2014–2026, ~ 756 games) using leave-one-tournament-out cross-validation, targeting a Brier score below 0.170 (vs. ~ 0.200 seed baseline and ~ 0.175 Vegas benchmark).

1 Introduction

The NCAA Division I Men’s Basketball Tournament is a 68-team, single-elimination event producing 63 games over three weeks each March. Its high variance—driven by single-elimination format, large talent gaps, and emotional intensity—makes it a canonical testbed for probabilistic prediction.

The Kaggle *March Machine Learning Mania* competition, running annually since 2014 with a \$50,000 prize pool, asks participants to submit $P(\text{Team}_i \text{ beats Team}_j)$ for every possible matchup, scored by Brier score:

$$\text{BS} = \frac{1}{N} \sum_{i=1}^N (p_i - o_i)^2 \quad (1)$$

where p_i is the predicted probability and $o_i \in \{0, 1\}$ is the outcome. Lower is better.

A key empirical finding from 12 years of competition: **simple models with strong features consistently outperform complex ML**. Most winners use logistic regression or Elo with KenPom efficiency ratings [Pomeroy, 2004]. The notable exception is Landgraf [2017], who won by modeling competitors’ submissions (meta-prediction) rather than improving game-level prediction.

This project aims to push the prediction frontier through structured model **heterogeneity, Transformer, Computer Vision, and market integration**, while respecting the fundamental constraint that 63 games per tournament provide very limited signal.

2 Datasets

2.1 Primary: Kaggle March ML Mania (2014–2026)

Twelve years of structured competition data comprising ~ 35 CSVs per edition. Table 1 summarizes the core files.

The competition has evolved through four eras (Table 2), with 24 total competition instances across community, Google Cloud-sponsored, and variant-scoring formats.

Table 1: Core Kaggle dataset files (M = Men’s, W = Women’s prefix).

File	Description
MRegularSeasonDetailedResults	Box scores: FGM/A, 3PM/A, FTM/A, OR, DR, Ast, TO, Stl, Blk, PF
MNCAATourneyDetailedResults	Tournament game box scores (same schema)
MNCAATourneySeeds	Seedings W01–Z16 per season
MMasseyOrdinals	Daily rankings from 100+ systems (KenPom, Sagarin, BPI, RPI, ...)
MNCAATourneySlots	Bracket structure mapping
MGameCities	Game locations (travel distance features)
MTeamCoaches	Coaching tenure and tournament experience

Table 2: Kaggle NCAA competition history.

Era	Years	Sponsor	Format
Community	2014–2017	Kaggle	Unified, Log Loss
Google Cloud	2018–2020	Google Cloud + NCAA	M/W split + Analytics track
Return	2021–2022	Kaggle	M/W split + Spread variant
Unified	2023–2026	Kaggle (\$50K)	Combined M+W, Brier Score

2.2 External Rating Systems

Table 3: External data sources for feature engineering.

Source	Key Metrics	Cost	Access
KenPom	AdjO, AdjD, AdjEM, AdjT, Luck, SOS	\$24.95/yr	kenpompy
Barttorvik	T-Rank, Barthag, adj. efficiency	Free	Web scraping
Massey Ratings	100+ system composite	Free	In Kaggle CSV
EvanMiya	BPR (Bayesian Performance Rating)	\$29.99/mo	Subscription
Sports Reference	Complete historical box scores	Free	sports-reference.com/cbb/

2.3 BigQuery Public Dataset

Google’s `bigquery-public-data.ncaa_basketball` dataset provides play-by-play data for every D1 game since 2009 (>40M plays), accessible via the free BigQuery tier.

2.4 Prediction Markets

Key market makers: SIG/Nellie Analytics (Kalshi’s first institutional MM), DRW (“Information Finance” desk), Jump Trading (equity stake in Kalshi), Jane Street (dedicated sports quant team).

3 Methodology

3.1 Feature Engineering

We construct features in four layers of increasing sophistication:

Layer 1: Static Team Ratings. Pre-tournament snapshots of KenPom AdjEM, Barttorvik T-Rank, Massey composite rank, seed number, and conference strength (avg. conference AdjEM).

Table 4: Prediction market platforms.

Platform	Maker Fee	API	Markets
Kalshi	0%	<code>docs.kalshi.com</code>	Championship, regional, game-by-game, props
Polymarket	Rebates	<code>py-clob-client</code>	450+ NCAA markets (non-US only)
DraftKings	—	Odds API	Spreads, totals, moneylines

Layer 2: Matchup Features. For each pair (A, B) : seed difference Δs , rating differences $\Delta r_{\text{POM}}, \Delta r_{\text{SAG}}, \Delta r_{\text{MOR}}$, tempo mismatch $|T_A - T_B|$, offensive–defensive style clash (O_A vs. D_B), geographic distance to game site, and coach tournament experience.

Layer 3: Temporal / Momentum. Rolling 10-game win percentage, average margin, late-season rating trend (Δr over last 30 days), and conference tournament momentum.

Layer 4: Market. Market-implied probability from Kalshi/Vegas, model–market disagreement signal $|p_{\text{model}} - p_{\text{market}}|$.

3.2 Model Architecture

We employ a three-tier model hierarchy (Figure ?? conceptual).

3.2.1 Tier 1: Baselines

1. **Seed Logistic:** $P(A \text{ wins}) = \sigma(\beta_0 + \beta_1 \cdot \Delta s)$ — the simplest meaningful model
2. **Elo Ratings:** Season-long Elo with tuned K -factor, home-court advantage, and inter-season regression
3. **KenPom Logistic:** Logistic regression on AdjEM difference — the standard strong baseline

3.2.2 Tier 2: Advanced Models

4. **Gradient Boosting** (XGBoost/LightGBM): Full feature set with walk-forward cross-validation. Expected to be the strongest single model based on competition history.
5. **Mixture-of-Experts (MoE):** The core methodological contribution. A gating network $g(\mathbf{x})$ classifies each matchup into K game types, and specialized expert models $f_k(\mathbf{x})$ handle each:

$$P(A \text{ wins} \mid \mathbf{x}) = \sum_{k=1}^K g_k(\mathbf{x}) \cdot f_k(\mathbf{x}) \quad (2)$$

Trained via EM: alternate between soft cluster assignment and expert parameter updates. Hypothesized game types:

- *Chalk* (1–4 vs. 13–16): High-seed dominance, rare upsets
 - *Competitive* (5–8 vs. 9–12): Upset-prone, seed provides weak signal
 - *Toss-up* (close seeds): Near coin-flip, momentum/matchup matters
 - *Cinderella* (mid-major vs. power): Conference mismatch dynamics
6. **Transformer Sequence Model:** Encode each team’s season as a sequence of game embeddings:

$$\mathbf{h}_{\text{team}} = \text{TransformerEncoder}(\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_T)[\text{CLS}] \quad (3)$$

where $\mathbf{e}_t$ encodes game t ’s opponent rating, margin, location, and box score. Matchup prediction via MLP on $[\mathbf{h}_A; \mathbf{h}_B]$. Architecture: 4 layers, 64-dim, 4 heads. Pretrained on regular-season games, fine-tuned on tournament outcomes.

3.2.3 Tier 3: Meta-Models

7. **Weighted Ensemble:** $\hat{p} = \sum_m w_m \hat{p}_m$, weights $\{w_m\}$ optimized on holdout Brier score via constrained optimization ($w_m \geq 0$, $\sum w_m = 1$).
8. **Market-Calibrated Ensemble:** Bayesian update of ensemble prediction $\hat{p}$ with market-implied probability p_{mkt} :

$$p_{\text{final}} = \frac{\hat{p}^\alpha \cdot p_{\text{mkt}}^{1-\alpha}}{\hat{p}^\alpha \cdot p_{\text{mkt}}^{1-\alpha} + (1 - \hat{p})^\alpha \cdot (1 - p_{\text{mkt}})^{1-\alpha}} \quad (4)$$

where $\alpha \in [0, 1]$ controls trust in model vs. market (tuned on backtest).

9. **Portfolio Optimizer** (Kaggle-specific): Generate N bracket portfolios maximizing $P(\text{top-}K \text{ finish})$ given ensemble probabilities and estimated competitor distribution [Landgraf, 2017].

3.3 Kalshi Trading Strategy

- **Signal:** Trade when $|p_{\text{model}} - p_{\text{market}}| > \tau$ (threshold tuned on backtest)
- **Sizing:** Fractional Kelly criterion ($f = 0.25 \cdot f_{\text{Kelly}}^*$) for bankroll safety
- **Execution:** Limit orders via Kalshi REST API (0% maker fee)
- **Risk:** Max 5% bankroll per game, max 20% total exposure

4 Evaluation

4.1 Backtest Protocol

Leave-one-tournament-out (LOTO) cross-validation on 12 tournaments (2014–2026). For each holdout season s :

1. Train on all seasons $\neq s$ (regular season + tournament)
2. Predict all tournament matchups in season s
3. Compute Brier score on ~ 63 games

Report mean $\pm$ std Brier across 12 folds.

4.2 Benchmarks

Table 5: Target Brier score benchmarks.

Model	Brier Score
Coin flip (0.5 for all)	0.250
Seed baseline	~ 0.200
KenPom logistic	~ 0.185
Vegas line	~ 0.175
Our target	< 0.170

5 Timeline

Table 6: Project timeline (6 weeks, March 20 – May 1, 2026).

Week	Date	Deliverable
1–2	Mar 20 – Apr 3	Data pipeline, EDA notebook, feature matrix
2–3	Apr 3 – Apr 10	Baseline models (Elo, Logistic, XGBoost), LOTO backtest
3–4	Apr 10 – Apr 17	MoE implementation + Transformer sequence model
4–5	Apr 17 – Apr 24	Full 12-year backtest, ensemble optimization
5–6	Apr 24 – May 1	Kalshi API integration, paper draft, Kaggle 2027 pipeline

6 Risk Assessment

Table 7: Key risks and mitigations.

Risk	Impact	Mitigation
63 games insufficient for DL	High	Heavy regularization; pretrain on regular season ($\sim 5,000$ games/yr)
KenPom paywall changes	Medium	Barttorvik (free) as primary fallback
Kalshi regulatory shut-down	Medium	Backtest-only mode; DraftKings as alternative
Simple models dominate	Medium	Always benchmark; value is in MoE insight, not raw Brier
Single-tournament luck	High	Evaluate on 12-year LOTO, not single tournament

References

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